

PDS PRACTICAL-7

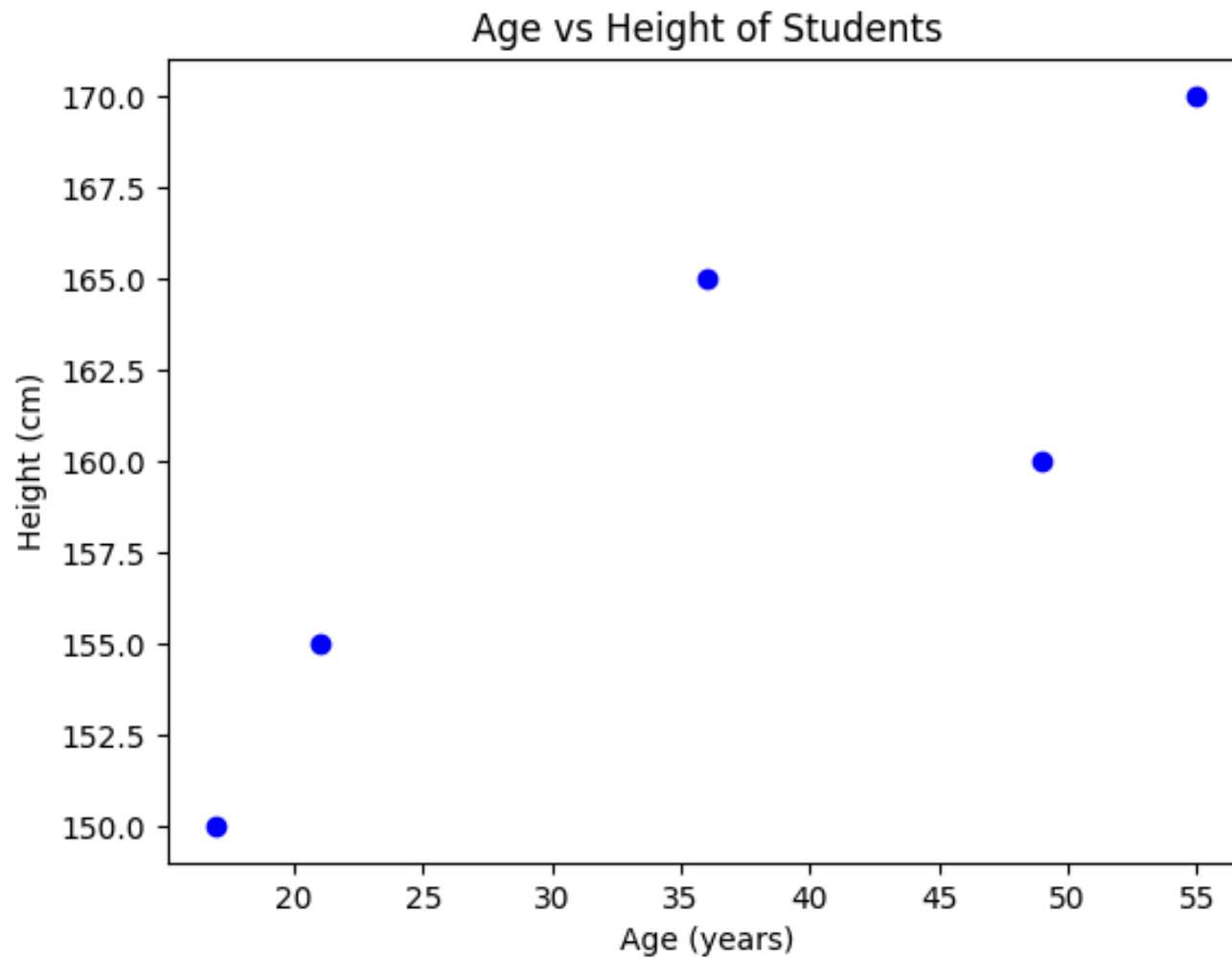
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Class :S.Y.I.T	Batch: 1
Date of Assignment : 2/09/2026	Date/Time of Submission : 2/09/2026

```
import matplotlib.pyplot as plt
```

a. Write a Python program to create a scatter plot of Age and Height of 5 students.

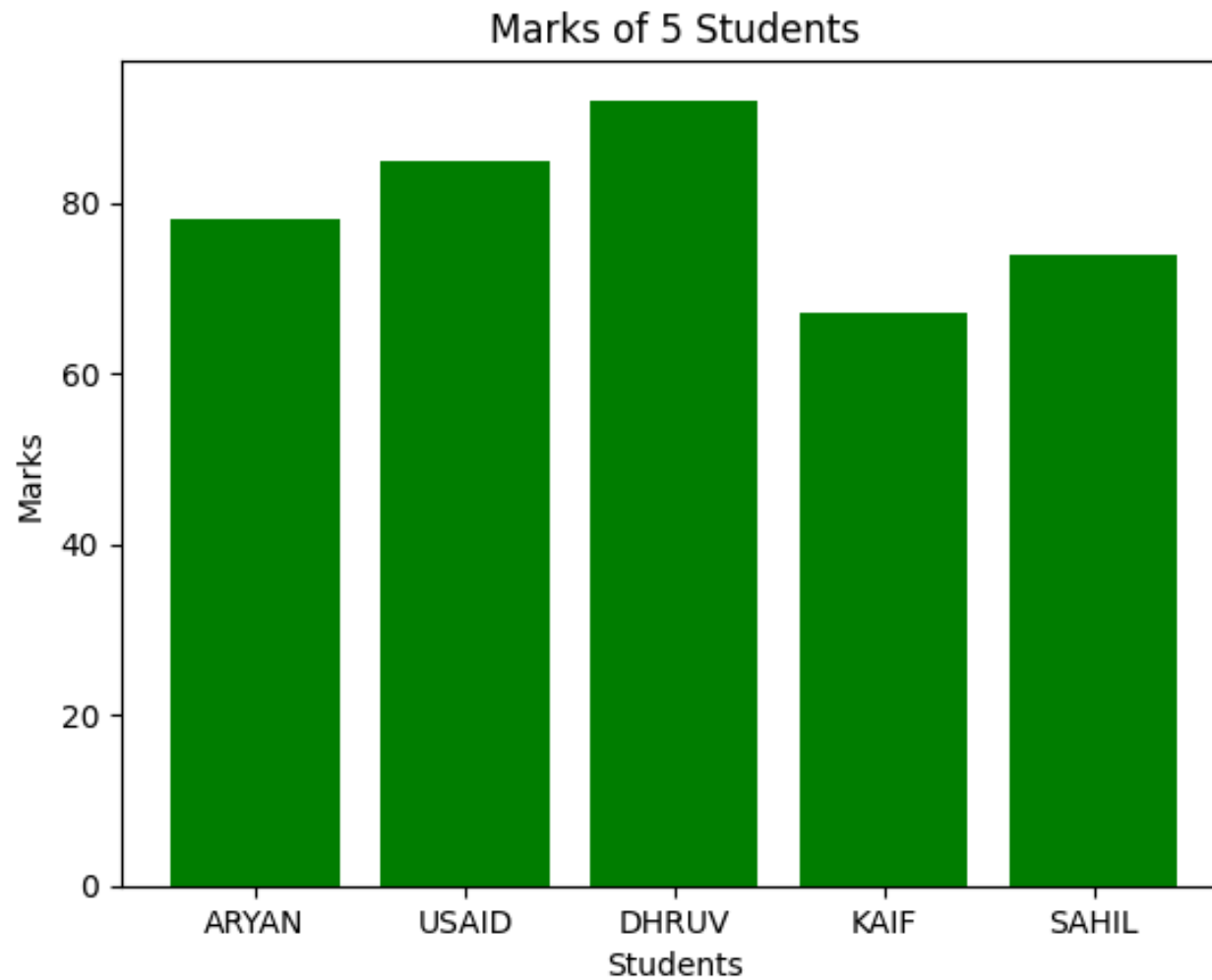
```
age = [17, 21, 49, 36, 55]
height = [150, 155, 160, 165, 170]

plt.scatter(age, height, color='blue')
plt.title('Age vs Height of Students')
plt.xlabel('Age (years)')
plt.ylabel('Height (cm)')
plt.show()
```



b. Write a Python program to create a bar chart showing marks of 5 students.

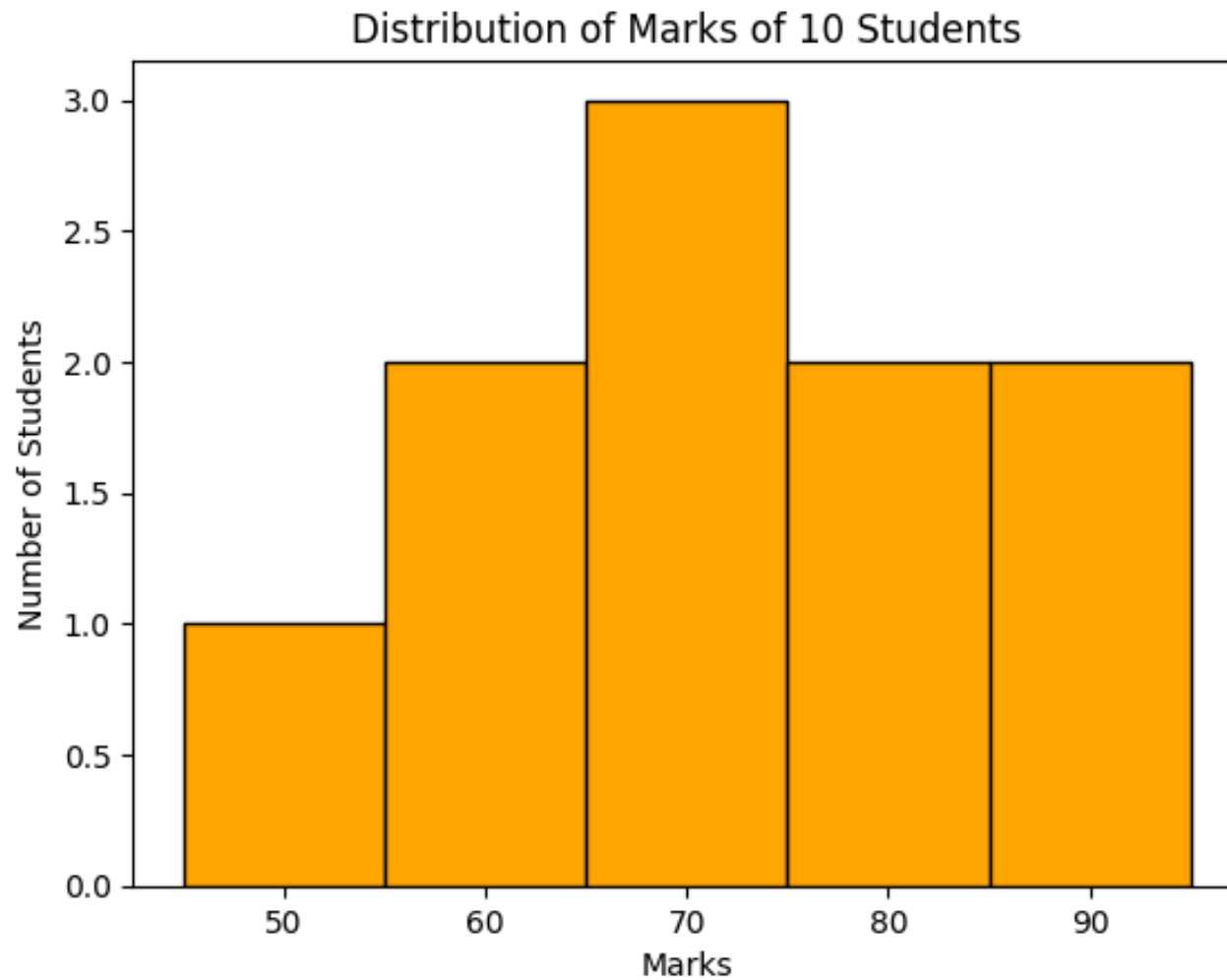
```
students = ['ARYAN', 'USAID', 'DHRUV', 'KAIF', 'SAHIL']  
marks = [78, 85, 92, 67, 74]  
  
plt.bar(students, marks, color='green')  
plt.title('Marks of 5 Students')  
plt.xlabel('Students')  
plt.ylabel('Marks')  
plt.show()
```



c. Write a Python program to create a histogram showing the distribution of marks of 10 students.

```
marks = [45, 55, 60, 65, 70, 72, 78, 80, 88, 95]

plt.hist(marks, bins=5, color='orange', edgecolor='black')
plt.title('Distribution of Marks of 10 Students')
plt.xlabel('Marks')
plt.ylabel('Number of Students')
plt.show()
```

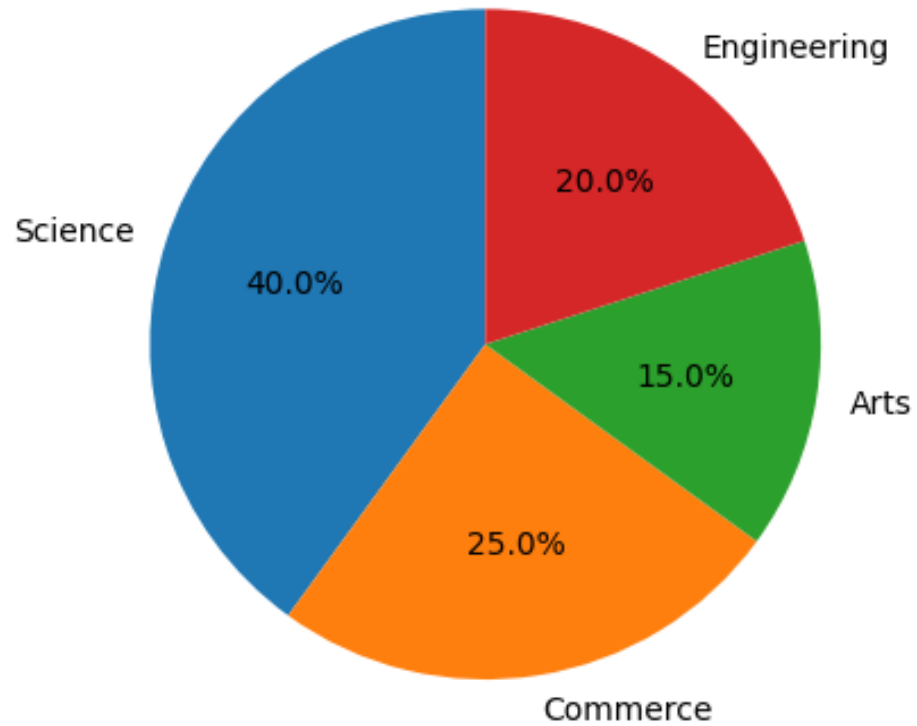


d. Write a Python program to create a pie chart showing the number of students in different courses.

```
courses = ['Science', 'Commerce', 'Arts', 'Engineering']
students = [40, 25, 15, 20]

plt.pie(students, labels=courses, autopct='%1.1f%%', startangle=90)
plt.title('Number of Students in Different Courses')
plt.show()
```

Number of Students in Different Courses



e. Write a Python program to create a bar chart showing sales of 5 different products.

```
products = ['Product A', 'Product B', 'Product C', 'Product D', 'Product E']
sales = [220, 340, 180, 410, 290]

plt.bar(products, sales, color='purple')
plt.title('Sales of 5 Different Products')
plt.xlabel('Products')
plt.ylabel('Sales (units)')
plt.show()
```

Sales of 5 Different Products

